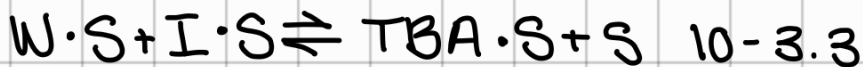
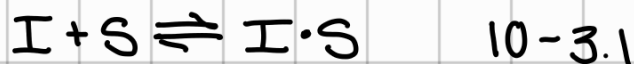


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P10-3:

rxn mech:



rate laws w/:

a) surface rxn is rate-limiting

+ two species absorbed = dual-site Langmuir

$$\text{norm} \rightarrow r_s = k_s \left(\frac{C_A \cdot s C_V - C_B \cdot s C_U}{k_s} \right)$$

required for TBA:

$$r_s = k_s \left(\frac{C_A \cdot s C_B \cdot s - C_C \cdot s C_S}{k_s} \right)$$

rate on surface:

$$r'_T = k_c \left(\frac{C_I s C_W s - C_T s C_S}{k_c} \right)$$

site balance (one-type):

$$C_t = C_s + K_I C_I C_s + K_W C_W C_s + K_T C_T C_s$$

$$C_t = C_s (1 + K_I C_I + K_W C_W + K_T C_T)$$

$$C_s = \frac{C_t}{(1 + K_I C_I + K_W C_W + K_T C_T)}$$

for $W \cdot S + I \cdot S \rightleftharpoons TBA \cdot S + S$ dual site

$$r_s = k_s \left(\frac{C_A s C_B s - C_C s C_S}{k_s} \right)$$

overall equili. constant:

$$K_c = \frac{C_T}{C_I C_W} \rightarrow C_I C_W - \frac{C_T}{K_c}$$

adsorption :

$$C_{IS} = K_I C_I C_S ; C_{WS} = K_W C_W C_S ,$$

$$C_{TS} = K_T C_T C_S$$

$$C_{WS} \cdot C_{IS} = (K_W C_W C_S)(K_I C_I C_S) = K_I K_W C_I C_W C_S^2$$

$$C_{TS} \cdot C_S = (K_T C_T C_S) C_S = K_T C_T C_S^2$$

$$r'_T = k_s C_S^2 \left(K_I K_W C_I C_W - \frac{K_T C_T}{K_S} \right)$$

$$C_S^2 = \frac{C_t^2}{(1 + K_I C_I + K_W C_W + K_T C_T)^2}$$

$$r'_T = \frac{k_s C_t^2}{(1 + K_I C_I + K_W C_W + K_T C_T)^2} \left(K_I K_W C_I C_W - \frac{K_T C_T}{K_S} \right)$$

factor :

$$K_I K_W C_I C_W - \left(\frac{K_T C_T}{K_S} \right) = K_I K_W \left(C_I C_W - \frac{C_T}{K_C} \right)$$

$$\therefore K_C = \frac{K_S K_T K_W}{K_I}$$

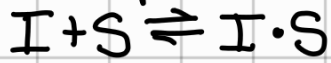
into kinetic constant :

$$k_a = k_s C_t^2 K_I K_W$$

plug back in :

$$-r'_I = r'_T \rightarrow = \frac{k_a (C_I C_W - C_T / K_C)}{(1 + K_I C_I + K_W C_W + K_T C_T)^2}$$

b) adsorption is limiting



$$\text{single-step: } r_{AD} = k_{AD} (C_I C_S - C_{IS} / K_A)$$

$$\therefore K_A = k_{AD} / k_{-AD}$$

for $I \rightleftharpoons S \rightarrow$

$$r'_T = -r'_I = r_{AD} = k_{AD} (C_I C_S - C_{IS} / K_I)$$

Surface rxn equil $\therefore$ solve for C_{IS} :

$$K_{sr} = \frac{C_{TS} C_S}{C_{ws} C_{IS}} \rightarrow C_{IS} = \frac{C_{TS} C_S}{K_{sr} C_{ws}}$$

$$C_{TS} = K_T C_T C_S \therefore C_{ws} = K_w C_w C_S$$

$$C_{IS} = \frac{K_I C_T C_S \cdot C_S}{K_{sr} K_w C_w C_S} \rightarrow = \frac{K_I}{K_{sr} K_w} \frac{C_T}{C_w} \cdot C_S$$

site balance (one-type from before)

w/ C_{ws} , C_{TS} $\therefore$ C_{IS} to get

$$C_t = C_s \left[1 + K_w C_w + K_T C_T + \frac{K_I}{K_{sr} K_w} \cdot \frac{C_T}{C_w} \right] \rightarrow$$

$$C_s = \left[\frac{C_t}{\left(1 + K_w C_w + K_T C_T + \frac{K_I}{K_{sr} K_w} \cdot \frac{C_T}{C_w} \right)} \right]^*$$

$$\frac{C_{IS}}{K_I} = \frac{K_T}{K_I K_{sr} K_w} \cdot \frac{C_T}{C_w} \cdot C_s$$

$$r'_T = k_{AD} C_s \left[C_I - \frac{K_T}{K_I K_{sr} K_w} \frac{C_T}{C_w} \right] \text{ w/ } C_s \rightarrow$$

$$-r'_T = \frac{k_{AD} C_t}{\left(1 + K_w C_w + K_T C_T + \frac{K_I}{K_{sr} K_w} \cdot \frac{C_T}{C_w} \right)} \cdot \left[C_I - \frac{K_T}{K_I K_{sr} K_w} \cdot \frac{C_T}{C_w} \right]$$

↓

$$C_I - \frac{k_T}{k_I k_{sr} k_w} \frac{C_T}{C_w} \rightarrow \frac{1}{C_w} \left(C_I C_w - \frac{C_T}{k_c} \right)$$

$$\therefore k_c = \frac{k_I k_{sr} k_w}{k_T} \rightarrow k_b = \frac{k_{ad} C_t}{C_w}$$

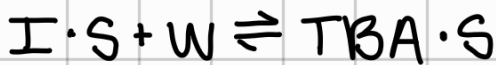
So new denominator: $\rightarrow$

$$D_b = 1 + k_w C_w + k_T C_T + \frac{k_T}{k_{sr} k_w} \frac{C_T}{C_w}$$

so $-r'_I / r'_T \rightarrow$

$$r'_T = \frac{k_b (C_I C_w - C_T / k_c)}{1 + k_w C_w + k_T C_T + \frac{k_T}{k_{sr} k_w} \frac{C_T}{C_w}}$$

c) 10-3.5 w/ surface rxn is limiting



fast adsorption: $C_{IS} = k_I C_I C_S$ $\therefore C_{TS} = k_T C_T C_S$

site balance:

$$C_t = C_S + C_{IS} + C_{TS} \rightarrow C_S (1 + k_I C_I + k_T C_T)$$

$$C_S = \frac{C_t}{(1 + k_I C_I + k_T C_T)}$$

surface rxn:

$$r_s = k_s \left(C_{IS} C_w - \frac{C_{TS} C_w}{k_s} \right) = r'_T \rightarrow$$

$$-r'_I = r'_T = k_s C_w \left(C_{IS} - \frac{C_{TS}}{k_s} \right)$$

sub $C_{IS} =$ $\therefore C_{TS} =$ into r'_T

$$r'_T = k_s C_w C_S \left(k_I C_I - \frac{k_T C_T}{k_s} \right) \rightarrow C_S \rightarrow$$

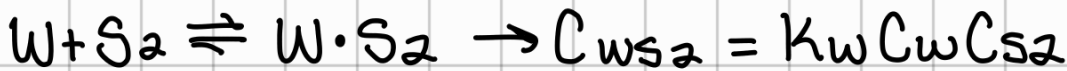
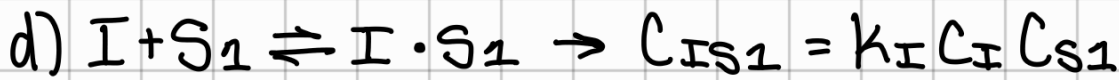
$$r'_T = \frac{k_s C_w C_t}{(1 + k_I C_I + k_T C_T)} \left(k_I C_I - \frac{k_T C_T}{k_s} \right)$$



$$k_I C_I - \frac{k_T C_T}{k_s} = k_I \left(C_I - \frac{C_T}{k_c} \right)$$

$$\therefore k_c = \frac{k_s k_I}{k_T} \rightarrow k_c = k_s k_I C_t \rightarrow$$

$$r'_T = \frac{k_c (C_I C_w - C_T / k_c)}{(1 + k_I C_I + k_T C_T)}$$



rate limiting step: $I \cdot S_1 + W \cdot S_2 \rightleftharpoons TBA + S_1 + S_2$
surface rxn:

$$r_s = k_s \left(C_{IS_1} C_{WS_2} - \frac{C_T C_{S_1} C_{S_2}}{k_{sr}} \right)$$

$$k_{sr} = \frac{C_T C_{S_1} C_{S_2}}{C_{IS_1} C_{WS_2}} @ \text{equili}$$

↳ surface rxn equili const.

site balances:

$S_1: C_{t1} = C_{IS_1} + C_{S_1}$ w/ $C_{IS_1} = k_I C_I C_{S_1} \rightarrow$
 $C_{t1} = C_{S_1} (1 + k_I C_I) \rightarrow C_{S_1} = \frac{C_{t1}}{1 + k_I C_I} *$

$S_2: C_{t2} = C_{WS_2} + C_{S_2}$ w/ $C_{WS_2} = k_W C_W C_{S_2} \rightarrow$
 $C_{t2} = C_{S_2} (1 + k_W C_W) \rightarrow C_{S_2} = \frac{C_{t2}}{1 + k_W C_W} *$

$$C_{IS_1} = k_I C_I \left(\frac{C_{t1}}{1 + k_I C_I} \right) \quad ; \quad C_{WS_2} = k_W C_W \left(\frac{C_{t2}}{1 + k_W C_W} \right)$$

$$C_{S_1} C_{S_2} = \left(\frac{C_{t1}}{1 + k_I C_I} \right) \left(\frac{C_{t2}}{1 + k_W C_W} \right) \rightarrow$$

$$= \left(\frac{C_{t1} C_{t2}}{(1 + k_I C_I)(1 + k_W C_W)} \right)$$

rate:

$$r'_T = k_s \left(C_{I51} C_{W52} - \frac{C_T C_{S1} C_{S2}}{k_{sr}} \right) \text{ sub in } C_{I51} \text{ and } C_{W52}$$

$$= k_s \left(k_I C_I \left(\frac{C_{t1}}{1 + k_I C_I} \right) \cdot \left(k_W C_W \left(\frac{C_{t2}}{1 + k_W C_W} \right) \right) - \dots \rightarrow$$

$$- \frac{C_T C_{t1} C_{t2}}{(1 + k_I C_I)(1 + k_W C_W) k_{sr}}$$

$$r'_T = \frac{k_s C_{t1} C_{t2}}{(1 + k_I C_I)(1 + k_W C_W)} \left(k_I k_W C_I C_W - \frac{C_T}{k_{sr}} \right)$$

$$B = k_I k_W C_I C_W - \frac{C_T}{k_{sr}} \quad \downarrow \quad k_I k_W \left(\frac{C_I C_W - \frac{C_T}{k_{sr}}}{k_{sr}} \right)$$

$$B = k_I k_W C_I C_W - k_I k_W \left(\frac{C_T}{k_{sr} k_I k_W} \right)$$

$$B = k_I k_W \left[C_I C_W - \left(\frac{C_T}{k_{sr} k_I k_W} \right) \right]$$

define $k_c = k_{sr} k_I k_W$

$$B = k_I k_W \left[C_I C_W - \frac{C_T}{k_c} \right]$$

$$r'_T = \frac{k_s C_{t1} C_{t2}}{(1 + k_I C_I)(1 + k_W C_W)} \left(k_I k_W \left[C_I C_W - \frac{C_T}{k_c} \right] \right)$$

$$r'_T = \frac{k_s C_{t1} C_{t2} k_I k_W}{(1 + k_I C_I)(1 + k_W C_W)} \left(C_I C_W - \frac{C_T}{k_c} \right)$$

$$k = k_s C_{t1} C_{t2} k_I k_W$$

$$r'_T = \frac{k (C_I C_W - C_T / k_c)}{(1 + k_I C_I)(1 + k_W C_W)} = r'_T B A$$

e) The numerator is always for stoichiometry and equilibrium $\rightarrow C_I C_W - \frac{C_I}{K_c}$

The denominator always $\rightarrow K_c$

reflects site coverage and it shows the mechanism and rate-limiting step

P10-10

a) rate law: 0 order MEK

$$r'_{\text{MEK}} = k \quad (0 \text{ order})$$

$$\bar{r} = 0.0525 = k$$

b) rxn mechanism: rate-limiting step

Strong adsorption: $\text{Bu} + \text{S} \rightleftharpoons \text{B} \cdot \text{S}$

rate-limiting surface rxn: $\text{B} \cdot \text{S} \xrightarrow{k_s} \text{MEK} + \text{H}_2 + \text{S}$

weak adsorption = MEK $\rightleftharpoons$ H₂

$$\text{Langmuir: } \theta_{\text{Bu}} = \frac{K_{\text{Bu}} P_{\text{Bu}}}{1 + K_{\text{Bu}} P_{\text{Bu}} + K_{\text{MEK}} P_{\text{MEK}} + K_{\text{H}_2} P_{\text{H}_2}}$$

if Bu strongly absorbed $\rightarrow \theta_{\text{Bu}} = 1$

surface rxn rate per mass of catalyst:

$$r'_{\text{MEK}} = k_s \theta_{\text{Bu}} \approx k_s \rightarrow$$

independent of partial pressure

c) Change conversion to 90-95% and redo = 70-75%

There may be some internal diffusion limitations

since the outer shell is only active. The rate is

$r'_{\text{eff}} < r'_{\text{intri}}$. Then if it's non-isothermal

then the dehydrogenation is endothermic,

which causes the temp to be lowered. This leads to a lower $k(T)$.

d) matlab

e) Used a 0-order kinetic model at high Bu partial pressures. Now using lower Bu pressures then the surface is only somewhat covered and the Langmuir form of $r'_{\text{MEK}} = \frac{k K_{\text{Bu}} P_{\text{Bu}}}{1 + K_{\text{Bu}} P_{\text{Bu}}}$

The question is: how would this change the predicted conversion profile ($x(w)$) in a packed bed vs. the 0-order model, and what would

have to be done to show which is the best choice. This is critical thinking since it starts with an assumption (B_u saturated the surface). Then, there is comparison of two options and its affects. Lastly, experiments/testing is needed to complete and comprehend the two rate laws. The question uses assumptions first that then lead to thinking deeper with comparisons and different options to be further chosen with experiments.

f) matlab